



# Evaluation of BigBacter for Bacterial Surveillance and Cluster Detection



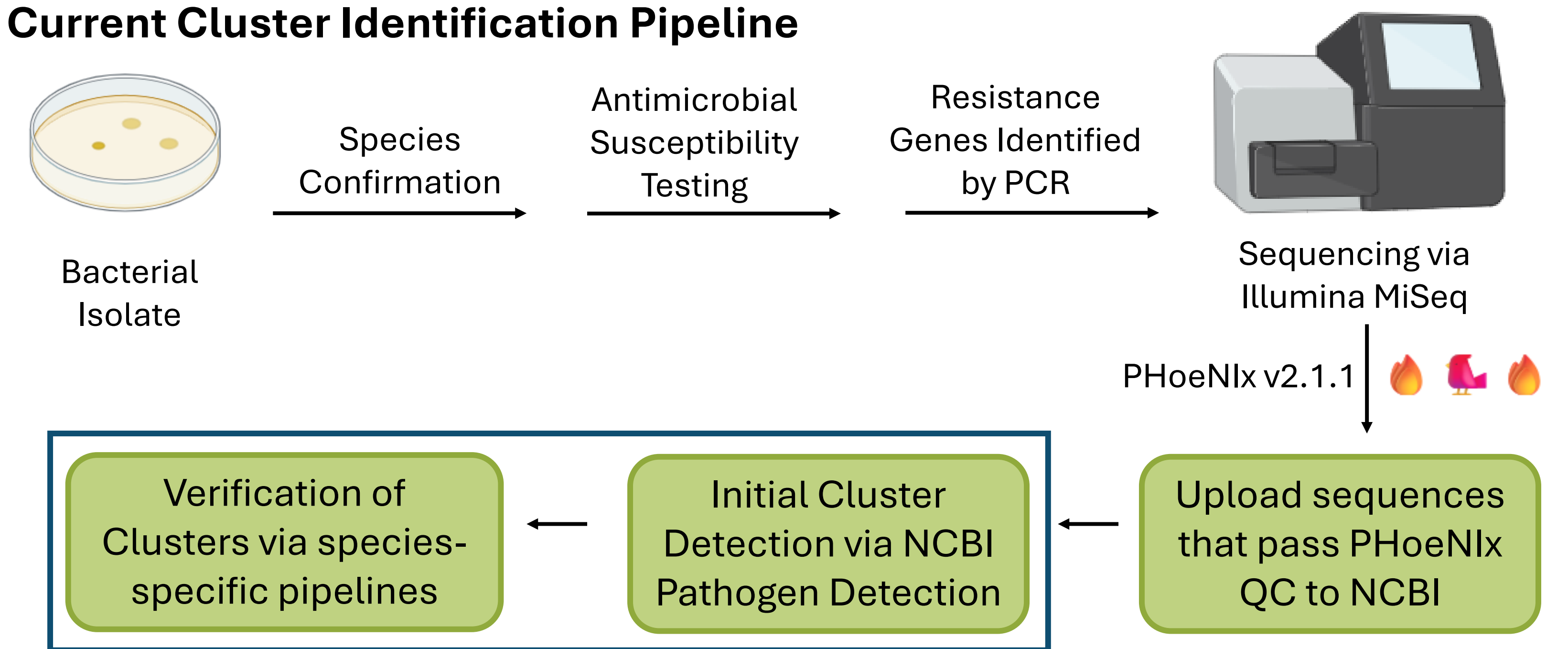
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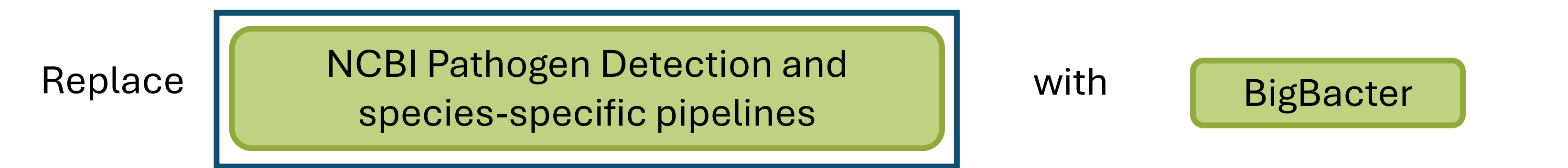
## Background

The Virginia Division of Consolidated Laboratory Services (DCLS) is the public health laboratory for the state of Virginia, responsible for performing bacterial pathogen genomic surveillance and reporting any identified genomic clusters to the Virginia Department of Health (VDH). Current methods for genomic surveillance and cluster detection are time intensive and reliant on external resources. DCLS aims to validate a single internal method that efficiently and accurately identifies genomic clusters for the many different bacterial species sequenced at DCLS.

### Current Cluster Identification Pipeline

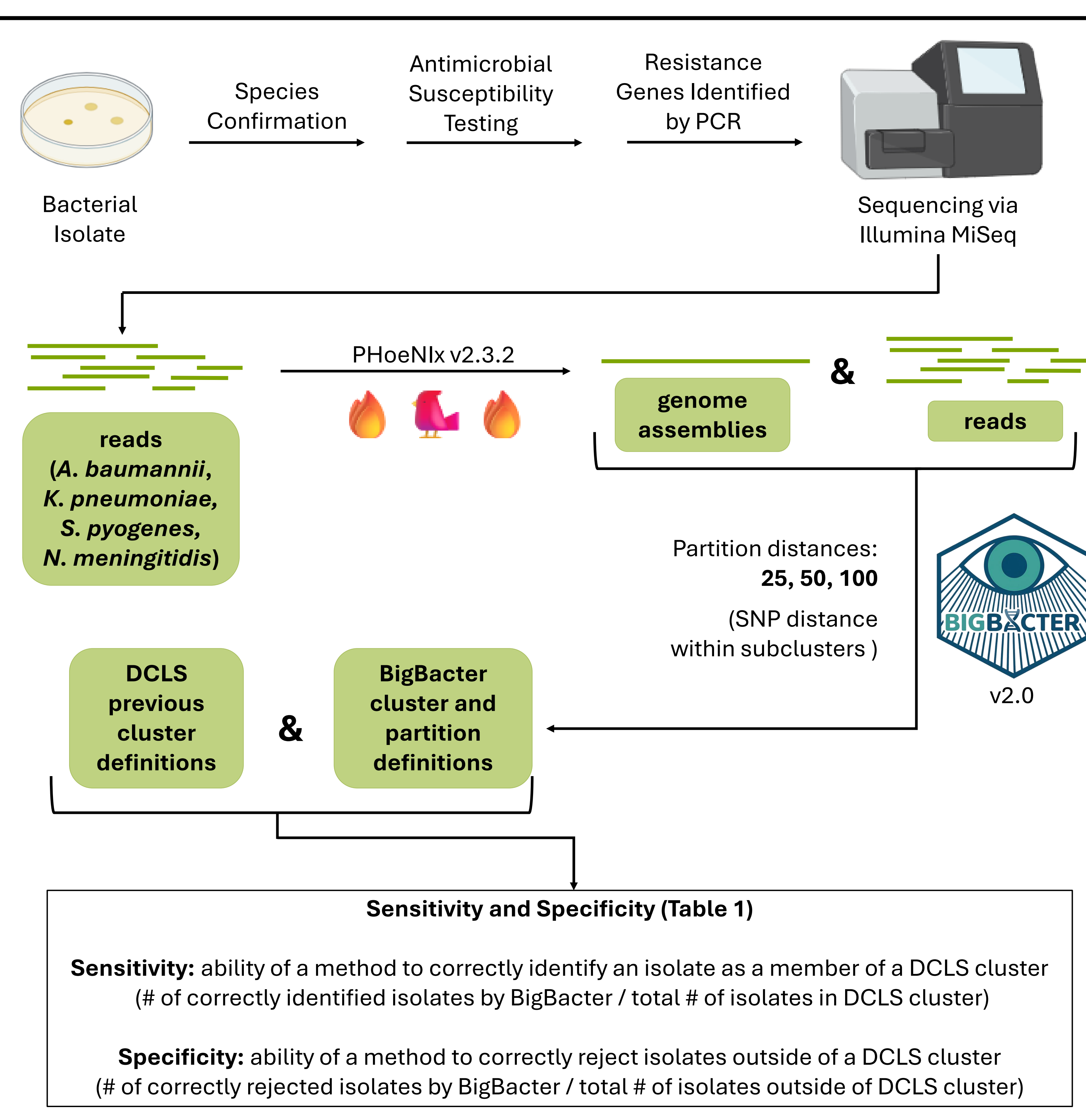


### Proposed Change



Towards this goal, BigBacter was tested for its potential to replace current methods—NCBI Pathogen Detection and species-specific comparative analysis pipelines, Dryad, Foushee, and BMGAP. Utilizing carbapenem-resistant *Acinetobacter baumannii* (CRAB) as an initial test case, BigBacter was evaluated on its ability to correctly assign each of 674 CRAB isolates to previously-identified outbreak clusters. Following this initial test on CRAB, BigBacter was also tested and evaluated on 51 *Neisseria meningitidis*, 278 *Streptococcus pyogenes*, and 186 carbapenem-resistant *Klebsiella pneumoniae* isolates.

## Methods



## Results

### Table 1. BigBacter Clustering Sensitivity and Specificity

**Main Finding: Partition distance of 25 was best for maximizing sensitivity and specificity across all species tested.**

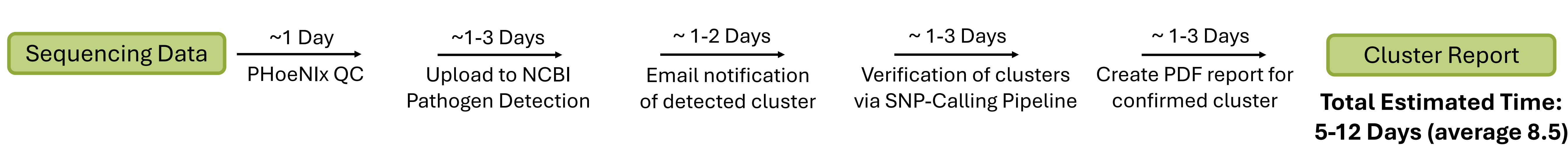
| Species (Isolates)                                        | Partition Distance (SNPs) | Cluster Average Sensitivity | Cluster Average Specificity | Cluster + Partition Average Sensitivity | Cluster + Partition Average Specificity |
|-----------------------------------------------------------|---------------------------|-----------------------------|-----------------------------|-----------------------------------------|-----------------------------------------|
| Carbapenem-resistant <i>Acinetobacter baumannii</i> (674) | 25                        | 1                           | 0.900959358                 | 1                                       | 0.932153664                             |
|                                                           | 50                        | 1                           | 0.900959358                 | 1                                       | 0.922715553                             |
|                                                           | 100                       | 1                           | 0.900959358                 | 1                                       | 0.919523976                             |
| Carbapenem-resistant <i>Klebsiella pneumoniae</i> (186)   | 25                        | 1                           | 0.955934921                 | 0.983333333                             | 0.996630300                             |
|                                                           | 50                        | 1                           | 0.955934921                 | 1                                       | 0.996630300                             |
|                                                           | 100                       | 1                           | 0.955934921                 | 1                                       | 0.982759414                             |
| <i>Neisseria meningitidis</i> (51)                        | 25                        | 1                           | 0.835282917                 | 0.906702899                             | 0.904289203                             |
|                                                           | 50                        | 1                           | 0.835282917                 | 0.956521739                             | 0.867130979                             |
|                                                           | 100                       | 1                           | 0.835282917                 | 1                                       | 0.835282917                             |
| <i>Streptococcus pyogenes</i> (278)                       | 25                        | 0.997864876                 | 0.933274485                 | 0.893869819                             | 0.988878306                             |
|                                                           | 50                        | 0.997864876                 | 0.933274485                 | 0.925857556                             | 0.981889633                             |
|                                                           | 100                       | 0.997864876                 | 0.933274485                 | 0.976655242                             | 0.937398855                             |

**Table 1.** Table depicting average sensitivity and specificity for each of four species at each of three partition distances. DCLS intends to move forward with the partition distance (within-subcluster SNP distance) that maximizes cluster sensitivity and cluster+partition specificity.

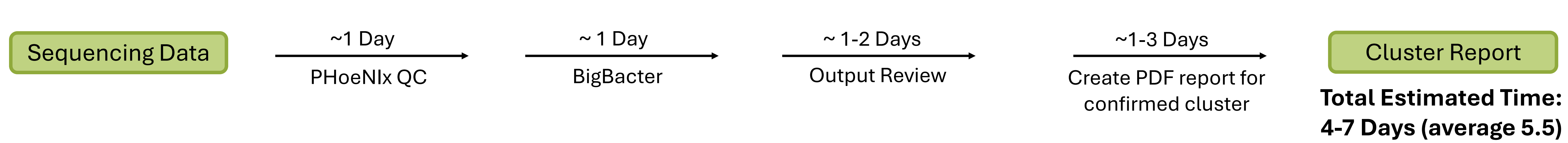
### Figure 2. BigBacter Efficiency

**Main Finding: BigBacter is expected to reduce turnaround time for cluster reporting in comparison to current methods**

#### A. Current Methods



#### B. BigBacter



**Figure 2.** Flowchart depicting (A) each step of current DCLS methods for genomic surveillance and cluster detection compared to (B) each step of the workflow when including BigBacter from sequencing data to PDF reports for the Virginia Department of Health (VDH). Time range estimates are provided for each step of each process, as well as the total average time each method is estimated to take from data to report.

## Discussion

**Main Finding: Partition distance of 25 was best for maximizing sensitivity and specificity across all species tested.**

For all species, there was no difference in cluster average sensitivity for the partition distances tested. For carbapenem-resistant *Klebsiella pneumoniae*, partition distances of 25 and 50 had the highest cluster+partition specificity. For carbapenem-resistant *Acinetobacter baumannii*, *Neisseria meningitidis*, and *Streptococcus pneumoniae*, using a partition distance of 25 resulted in the highest cluster+partition specificity. Overall, BigBacter's clustering performance was highly accurate when compared to previous DCLS cluster classifications, for all species tested.

**Main Finding: BigBacter is expected to reduce turnaround time for cluster reporting in comparison to current methods**

According to DCLS time estimates, BigBacter will take less time than current methods to get to cluster reports, on average. Even at BigBacter's longest estimated time of 7 days, it will take less time to achieve cluster reports than the average time taken by current methods. BigBacter's efficiency increase is also likely an underestimate, as DCLS currently requires different workflows for different species.

**Summary: DCLS will move forward with validating and implementing BigBacter for genomic surveillance and cluster detection**

From DCLS' analyses, BigBacter is both accurate and efficient. Replacing current methods with BigBacter would greatly improve genomic surveillance and cluster detection in Virginia. Validating a new, internal method would also reduce reliance on external resources such as NCBI. Following further validation and implementation of BigBacter, DCLS scientists will be able to report accurate genomic clusters of bacteria to VDH faster. This will ultimately lead to a better-informed public health response and reduced spread of these harmful pathogens in Virginia.

**Supplementary Materials & References**

